# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Beyond Earth** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

**(a)** 8 light-seconds **(b)** 8 light-years **(c)** 8 light-minutes **(d)** 8 light-hours

**2.** Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

**(a)** 10 kg **(b)** 360 kg **(c)** 6 kg **(d)** 60 kg

**3.** Which of the following explains day and night on Earth? 1

**(a)** the Moon blocks the Sun daily **(b)** Earth revolves around the Sun once a day
**(c)** the Sun moves around the Earth **(d)** Earth rotates on its axis

**4.** A student draws the Moon at the same time each night and sees the bright part grows. The Moon is: 1

**(a)** new **(b)** missing **(c)** waning **(d)** waxing

**5.** The weight of the same object on the Moon compared with Earth is: 1

**(a)** equal **(b)** one sixth **(c)** six times **(d)** zero

**6.** Which observation suggests that the Earth rotates? 1

**(a)** the Moon is round
**(b)** stars appear to move across the night sky
**(c)** the Sun is hot
**(d)** planets are bright

**7.** If the new moon occurs on 1 March, the next new moon will be about: 1

**(a)** 22 March **(b)** 8 March **(c)** 15 March **(d)** 30 March

**8.** If a satellite takes 90 minutes for one orbit, the number of orbits in 24 hours is: 1

**(a)** 8 **(b)** 90 **(c)** 16 **(d)** 24

**9.** **Assertion (A):** The Sun is a star. 1
**Reason (R):** The Sun produces its own light and heat.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** **Assertion (A):** The Moon does not produce its own light. 1
**Reason (R):** We see the Moon because it reflects sunlight.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** On a map of the solar system, 1 cm shows 150 million km. If the Sun-Earth distance is 150 million km, how far away is Mars at about 228 million km? 2

**12.** Why do planets appear to move against the background of stars? 2

**13.** Why is a launch near the equator more efficient for satellites? 2

**14.** A satellite goes round the Earth once in 100 minutes. How many orbits are completed in one day? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A day on planet X is 10 hours and its year is 5000 hours. How many days are there in its year? 3

**16.** Explain how the Sun's gravity keeps planets in orbit and what would happen if it vanished. 3

**17.** Explain how the first quarter Moon is seen from the Earth. 3

OR

Moon takes about 27 days to orbit the Earth. How many orbits in 1 year (365 days)? 3

**18.** Why do some comets appear only once in many years? 3

OR

Why are craters on the Moon preserved for long while Earth's are not? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** On a clear night in Shillong, Bhavya showed her younger brother the group of seven bright stars that makes a pattern like a big question mark, which people call the Saptarishi. She also pointed to a star that stays almost fixed in the northern sky and helps travellers find north. 5
(i) What do we call a group of stars forming a pattern?
(ii) What is the star that helps us to find north called?
(iii) Name the pattern of stars that Bhavya showed.
(iv) Why do stars appear to twinkle?

**20.** A student records the Moon's shape for 30 days and finds it was a thin crescent on day 3, half on day 8, full on day 15, half on day 22 and none on day 29. Analyse the pattern. 5

OR

Discuss the technological spin-offs of space research for daily life and health. 5

## END OF QUESTION PAPER — Total: 40 Marks